

Theory Questions

Basic Questions

1. What is a Servlet?
2. What are the advantages of Servlet over CGI?
3. What is the lifecycle of a Servlet?
4. What is HttpServlet?
5. What is web.xml?

Concept-Based Questions

6. Difference between GenericServlet and HttpServlet?
7. What are doGet() and doPost()?
8. What is request and response object?
9. What is RequestDispatcher?
10. Difference between forward() and sendRedirect()?

Exception Handling Questions

11. What is ServletException?
12. What is IOException in Servlet?
13. How to handle exceptions in servlet?
14. What is error-page in web.xml?
15. What happens if exception is not handled?

Advanced Questions

16. What is session management?
17. What are cookies?
18. What is HttpSession?
19. What is URL rewriting?
20. What is hidden form field?

Scenario-Based Questions

21. How do you handle login using servlet?
22. How do you track user session?
23. How do you pass data between servlets?
24. How do you handle multiple users?
25. How do you secure session data?

MCQ Questions

1. Servlet belongs to:
A. java.sql
B. javax.servlet
C. java.io
D. java.net

2. Lifecycle method called first:
A. service()
B. init()
C. destroy()
D. doGet()

3. Which handles GET request?
A. doPost()
B. doGet()
C. service()
D. init()

4. Which handles POST request?
A. doGet()
B. doPost()
C. destroy()
D. init()

5. Which object sends response?
A. Request
B. Response
C. Session
D. Cookie

6. Which is used to redirect?
A. forward()
B. sendRedirect()
C. include()
D. dispatch()

7. Session is used for:
- A. DB
 - B. User tracking
 - C. File handling
 - D. Streams
8. Cookie is stored in:
- A. Server
 - B. Client
 - C. DB
 - D. JVM
9. Which is used to forward request?
- A. RequestDispatcher
 - B. Response
 - C. Cookie
 - D. Session
- 10.** Which method destroys servlet?
- A. init()
 - B. destroy()
 - C. service()
 - D. run()

Programming MCQs

1. `req.getParameter("name");`

- A. Get input
- B. Set input
- C. Error
- D. Delete

2. `res.getWriter().println("Hello");`

- A. Output
- B. Input
- C. Error
- D. Session

3. `req.setAttribute("x",10);`

- A. Store data
- B. Get data
- C. Error
- D. Delete

4. `req.getAttribute("x");`

- A. Retrieve data
- B. Store data
- C. Error
- D. Delete

5. `HttpSession session = req.getSession();`

- A. Create session
- B. Delete session
- C. Error
- D. DB

6. `session.invalidate();`

- A. Create
- B. Destroy
- C. Error
- D. Print

7. `res.sendRedirect("home.jsp");`

- A. Forward
- B. Redirect
- C. Error
- D. Include

8. `RequestDispatcher rd = req.getRequestDispatcher("a.jsp");
rd.forward(req, res);`

- A. Redirect
- B. Forward
- C. Error
- D. Include

9. `Cookie c = new Cookie("user", "John");
res.addCookie(c);`

- A. Store cookie
- B. Delete cookie
- C. Error
- D. Session

10. `req.getRequestDispatcher("a.jsp").include(req, res);`

- A. Include
- B. Redirect
- C. Error
- D. Delete

Practice Programs

1. Simple Servlet Program

Description:

Create a basic servlet that displays a message in the browser.

Output:

"Hello, Servlet!"

2. Handle GET Request

Description:

Create a servlet that handles HTTP GET request and displays a response.

Task:

Use `doGet ()` method to process request

3. Handle POST Request

Description:

Create a servlet that handles HTTP POST request.

Task:

Use `doPost ()` method to process form data

4. Form Data Processing

Description:

Create an HTML form and process user input in a servlet.

Task:

- Accept input (name, age)
- Display entered values

5. Session Management

Description:

Maintain user session using HttpSession.

Task:

- Store username in session
- Retrieve and display it

6. Cookie Example

Description:

Store and retrieve user data using cookies.

Task:

- Create cookie
- Read cookie value

7. Login System

Description:

Develop a simple login system using servlet.

Task:

- Accept username/password
- Validate (hardcoded or DB)

Output:

Login Success / Login Failed

8. Forward vs Redirect Demo

Description:

Demonstrate difference between forward and redirect.

Task:

- Use `forward()` and `sendRedirect()`
- Show behavior difference

9. Multiple Servlet Communication

Description:

Pass data from one servlet to another.

Task:

- Use `RequestDispatcher`
- Transfer request/response

10. Logout System

Description:

Implement logout functionality.

Task:

- Invalidate session
- Redirect to login page